

FLYTBASE VISUAL INTELLIGENCE HACKATHON

State of the Art: Video Anomaly Detection

60 MIN · DEMOS AT THE END

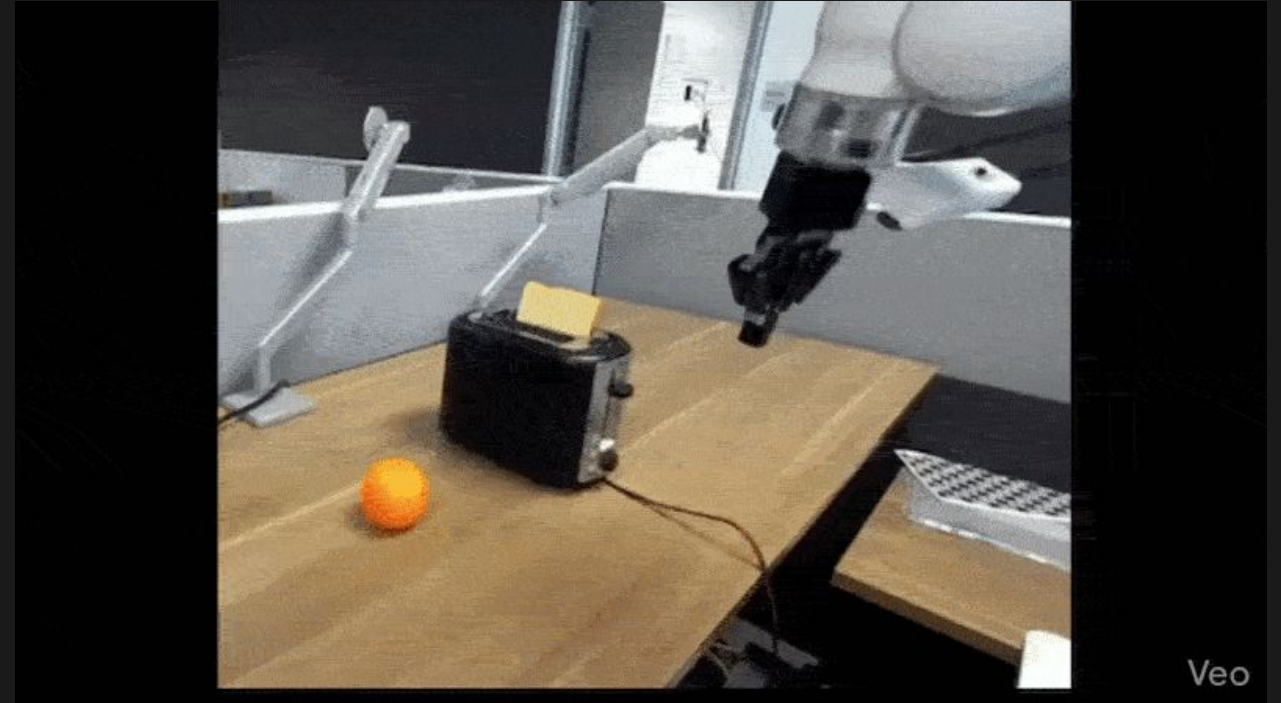
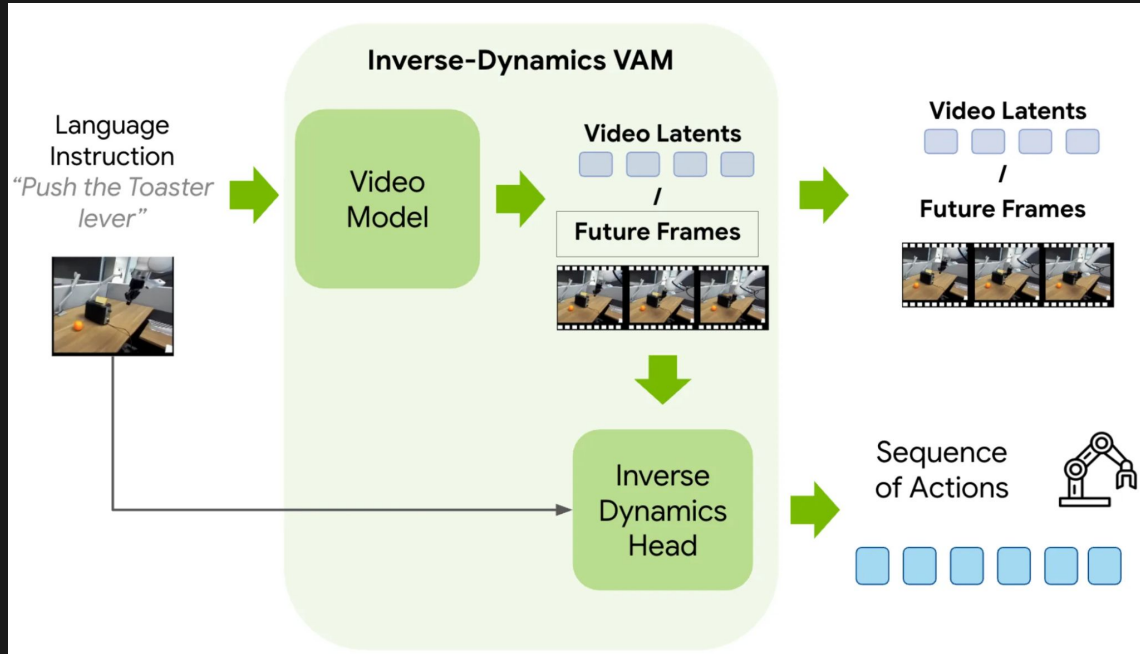
GPT 6 Astra



Robotics and its GPT 3 Moment



The Rise of World Action Models



THE EVIDENCE THAT MADE PEOPLE LOOK

On the April 2026 RoboArena leaderboard DreamZero scored 1750 against Pi-0.5 at 1622 — trained only on DROID, with no large cross-embodiment stage

Qwen3.6 Model and capabilities around

<https://qwen.ai/blog?id=qwen3.6>

SECTION 02

State of the art around VAD

Video anomaly detection — the papers, the approaches

The field renamed the problem twice in two years

Each rung adds an output the rung below could not produce.

VAD	Video Anomaly Detection A score per frame. Is something wrong, and when? No account of what.	2018–2024
VAU	Video Anomaly Understanding Detection plus description and explanation. The umbrella term for the whole output set.	2024–2025
VAR	Video Anomaly Reasoning Explicit multi-stage reasoning before the answer. Perception, then cognition, then verdict.	2025–2026

Cosmos-Embed1-448p-anomaly-detection

Detection by retrieval: text and video land in one vector space, and the anomaly is whatever ranks highest.

01 ENCODE BOTH SIDES

A five-second clip goes through the video tower and becomes a 768-dim vector. The sentence 'wrong-way driving' goes through the text tower into that same space.



02 NO CLASSIFIER HEAD

Nothing predicts a label. The LoRA fine-tune only pulled clip vectors closer to the phrases that describe them

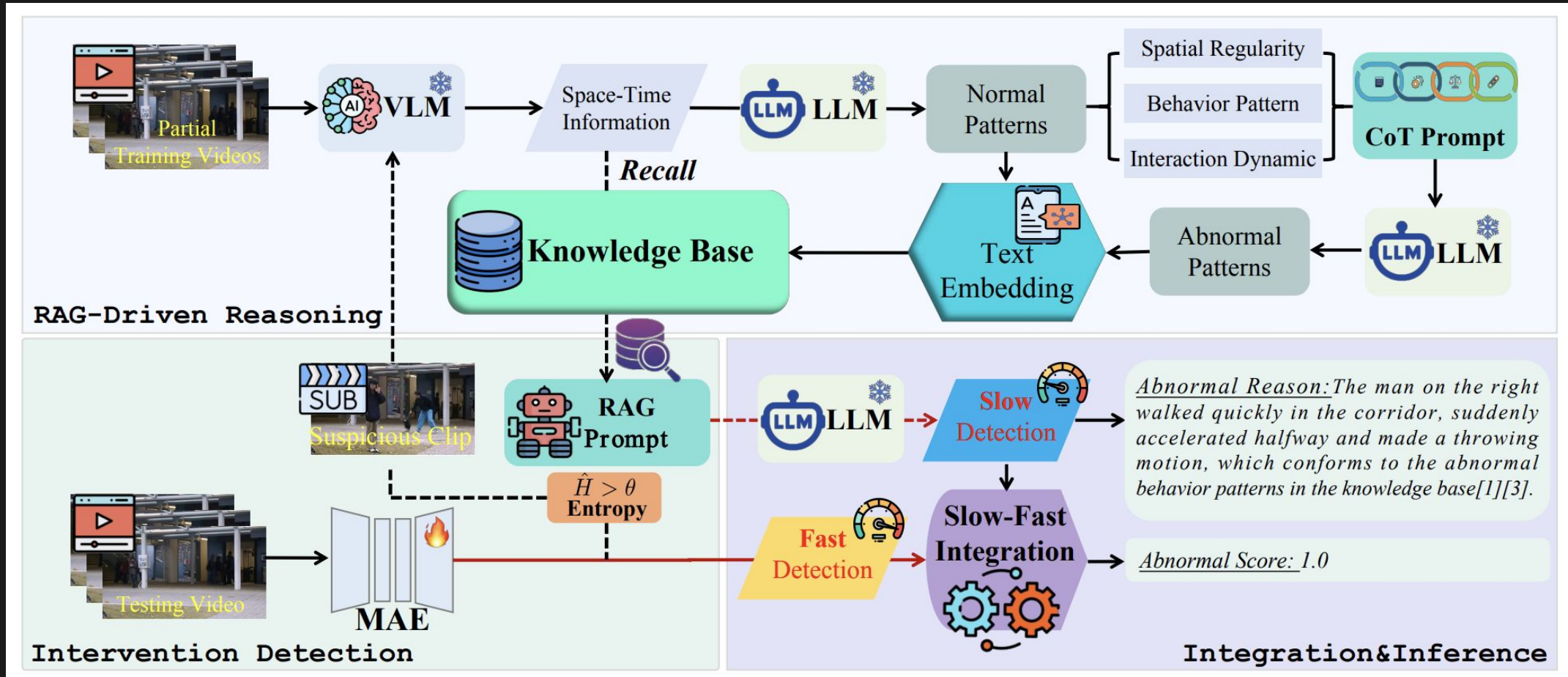


03 DETECT BY RANKING

Score every clip by cosine similarity to your query text. Sort. The top-k are your detections, and the query you typed is your class list.

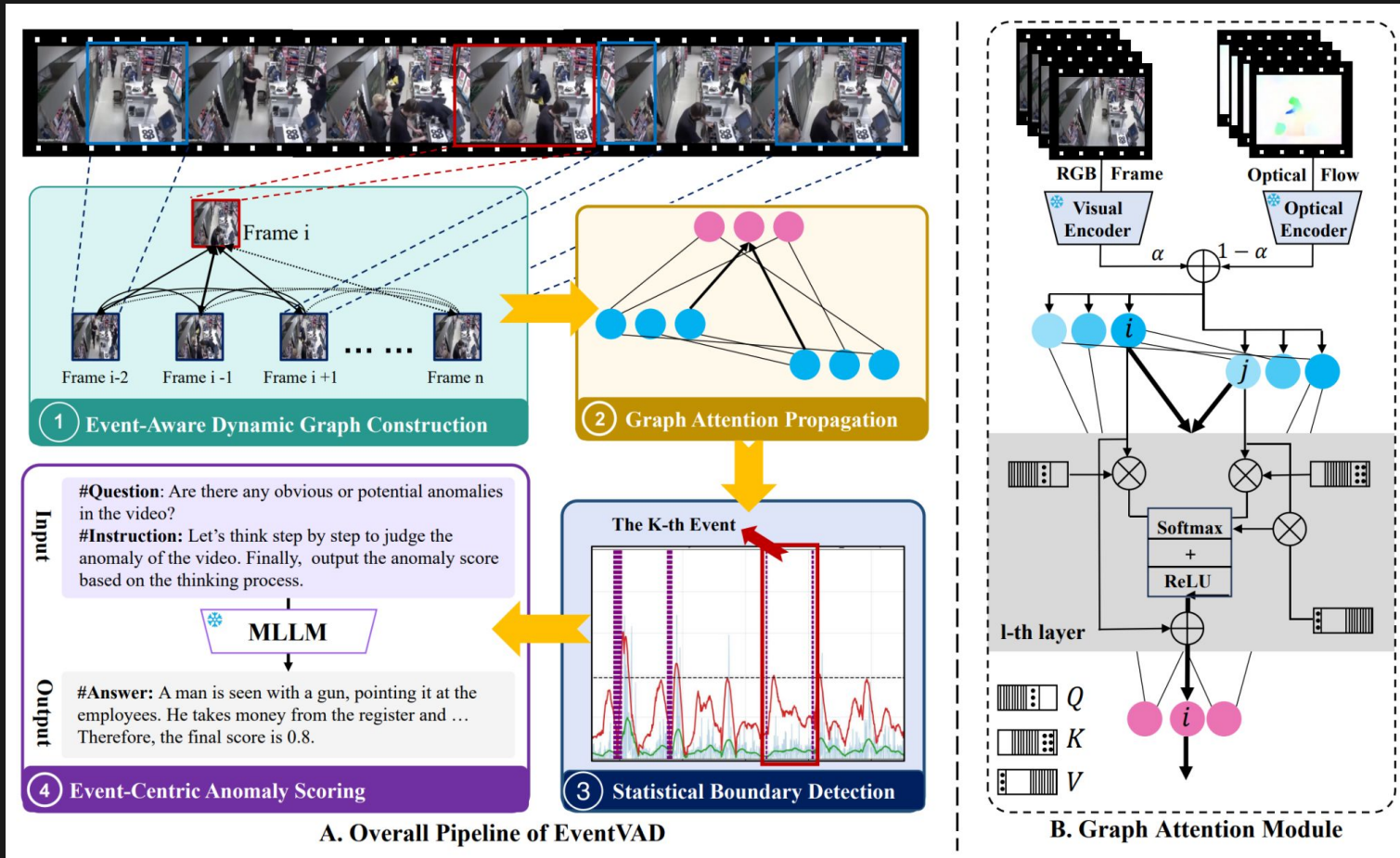
SlowFastVAD

A fast detector and a RAG-enhanced VLM, fused.



EventVAD – Training Free VAD

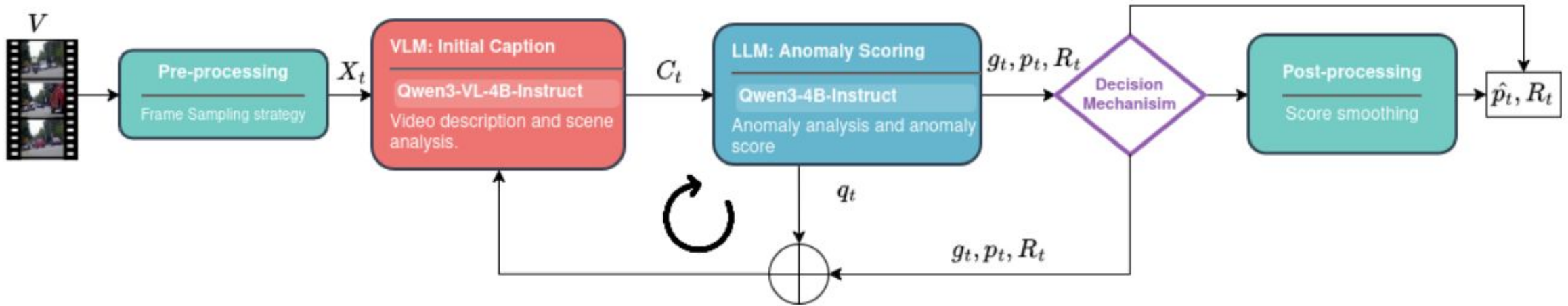
Training-free. Cut at event boundaries, then let the MLLM score.



QVAD

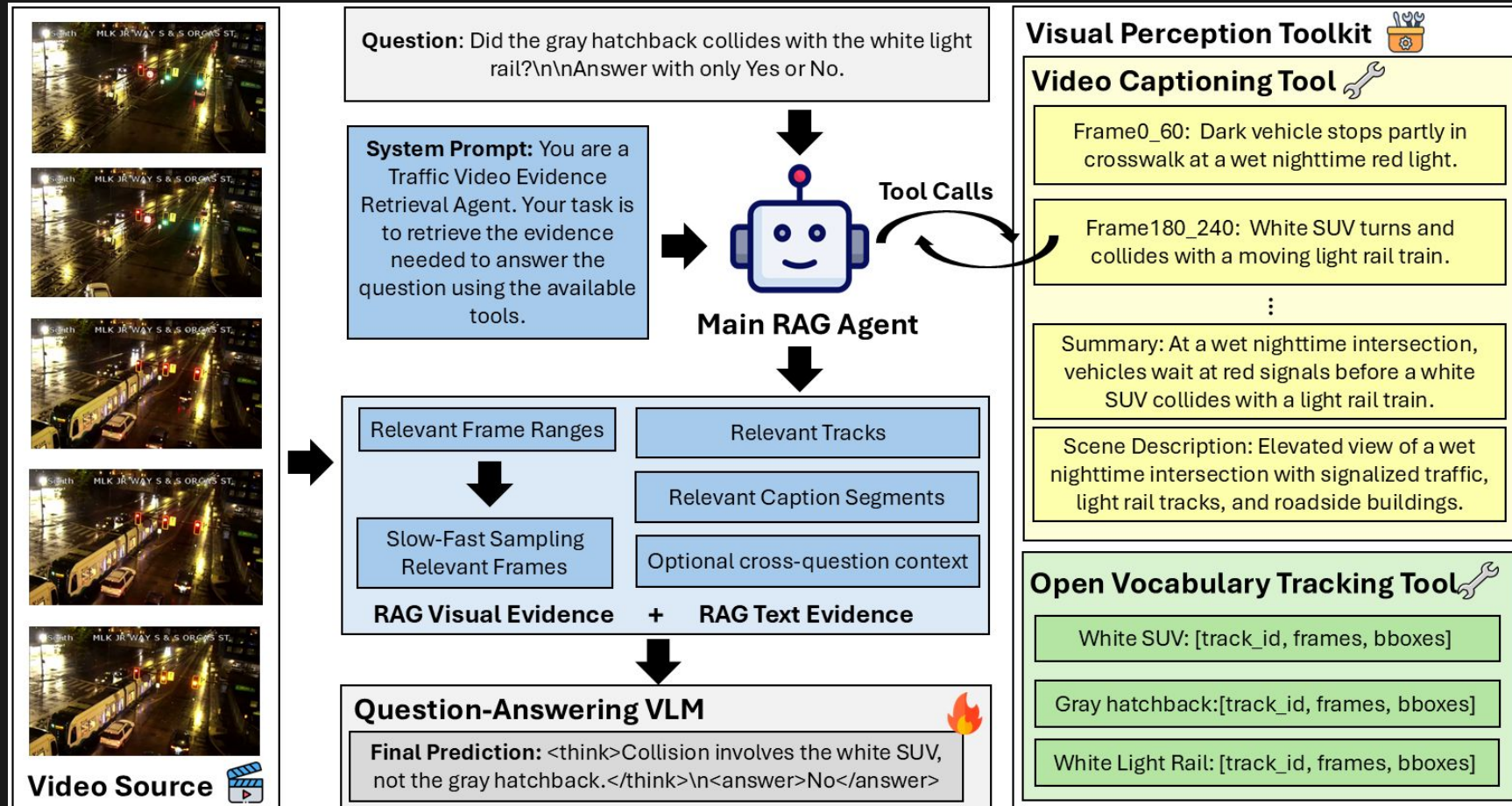
The bottleneck is not model capacity. It is the static question.

QVAD: Question-Centric Agentic Framework for Video Anomaly Detection



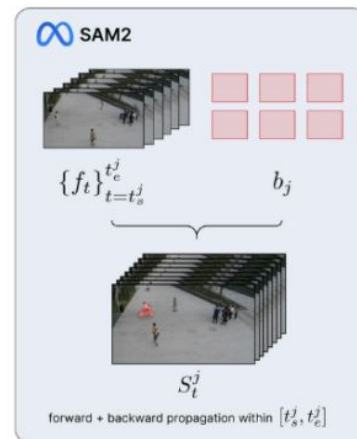
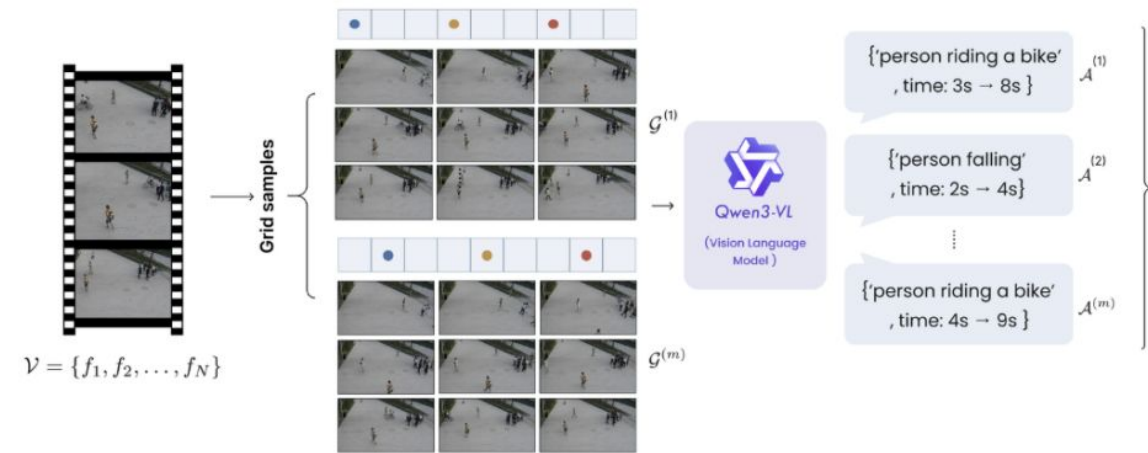
TAU-Agent

Agentic retrieval-augmented traffic anomaly understanding. AI City Challenge 2026.

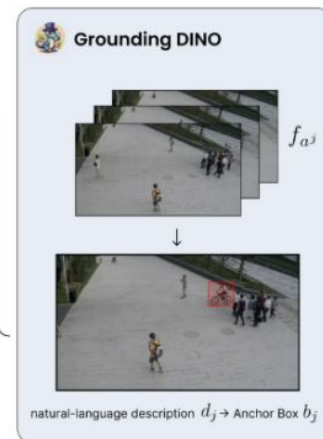


GridVAD

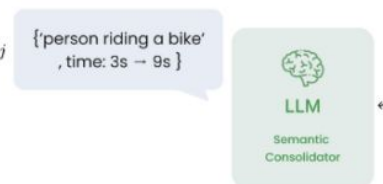
1) Grid Sampling & Anomaly Proposal



4) Temporal Propagation



3) Spatial Grounding



2) Self-Consistency Consolidation

AI City Challenge 2026 – Track 3

Table 3: Track 3 leaderboard entries connected to accepted papers.

Rank	Team ID	Team	Mean	Paper
1/27 (1/76)	25	Stellarview AI	0.6788	[26]; Qwen 3.5
- (2/76)	45	UOB&UW Team	0.6779	[34]; Qwen3VL-8B
2/27 (5/76)	60	FPT AI Vision	0.6703	[20]; Qwen3-VL-8B
3/27 (7/76)	12	Smart Vision	0.6669	[57]; Qwen
10/27 (26/76)	30	UWIPL ETRI	0.6185	[52]; Qwen
15/27 (36/76)	122	OptimAI	0.5880	[17]; Qwen
16/27 (41/76)	139	MR-CAS	0.5780	[31]; GPT
24/27 (55/76)	277	Korea Drive	0.4256	[25]; Qwen3

THE PAPER TO LOOK INTO

The 10th AI City Challenge - <https://arxiv.org/pdf/2608.17044>